

1. Administrative & Study Identification

Full Study Title: A Phase II, Multicenter, Randomized, Double Masked, Active Comparator-Controlled Study to Investigate the Efficacy, Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of R07200220 in Combination With Ranibizumab Administered Intravitreally in Patients With Diabetic Macular Edema **Short Title/Acronym:** BP43464 **Protocol Number:** BP43464 (EudraCT: 2021-004390-31) **NCT Number:** NCT05151744 **Sponsor Name:** Hoffmann-La Roche (Roche) **Investigational Product:** Vamikibart (R07200220) and Ranibizumab (Active Comparator; Trade Names: Lucentis, Susvimo, Cimerli, Byooviz; ATC Code: S01LA04) **Phase of Development:** Phase II **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To assess the efficacy, safety, tolerability, pharmacokinetics (PK), and pharmacodynamics (PD) of vamikibart (an IL-6 inhibitor) in combination with the anti-vascular endothelial growth factor (VEGF) inhibitor ranibizumab compared with ranibizumab alone in participants with diabetic macular edema (DME). * **Secondary Objectives:** To evaluate long-term visual acuity improvements (BCVA), reductions in central retinal thickness, and to monitor the safety profile over the extended observational period.

2.2 Background & Rationale Diabetic macular edema (DME) is a leading cause of vision loss in diabetic patients, driven by both vascular endothelial growth factor (VEGF) and inflammatory cytokines like interleukin-6 (IL-6). To address the gap in optimal visual recovery seen in some patients treated with anti-VEGF monotherapy alone, this study evaluates whether simultaneously inhibiting IL-6 (via vamikibart) and VEGF (via ranibizumab) provides superior efficacy in reducing macular edema and improving visual outcomes. Only one eye will be chosen as the study eye.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator-controlled (Ranibizumab + Sham) * **Blinding:** Double-masked (Double-blind) * **Randomization:** Randomized, Parallel Assignment * **Trial Status:** COMPLETED (Study has been completed)

3.2 Planned Enrollment & Duration * Target \$\$\$: 187 participants
* **Number of Sites:** 38 locations * **Estimated Study Start:** 17-Dec-2021 * **Estimated Primary Completion:** 17-Apr-2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of diabetes mellitus (Type 1 or Type 2). 2. Macular thickening secondary to diabetic macular edema (DME) involving the center of the macula. 3. Decreased visual acuity attributable primarily to DME in the study eye. 4. Ability and willingness to provide written informed consent and to comply with the study protocol. 5. Willingness to allow Aqueous Humor collection. 6. For women of childbearing potential: agreement to remain abstinent or use at least one highly effective contraceptive method that results in a failure rate of <1% per year during the treatment period and for at least 12 weeks after the final dose of study treatment.

4.2 Exclusion Criteria 1. Hemoglobin A1c (HbA1c) > 12%. 2. Uncontrolled blood pressure, defined as a systolic value > 180 mmHg and/or a diastolic value > 100 mmHg while a patient is at rest. 3. Currently pregnant or breastfeeding, or intend to become pregnant during the study. 4. Prior treatment with panretinal photocoagulation or macular laser to the study eye. 5. Any intraocular or periocular corticosteroid treatment within the past 16 weeks prior to Day 1 to the study eye. 6. Prior Iluvien or Retisert implants within 3 years prior to Day 1 to the study eye. 7. Prior or concomitant treatment with anti-VEGF therapy within 8 weeks prior to Day 1 to the study eye; Vabysmo within 16 weeks prior to Day 1, prior Beovu is not permitted. 8. Prior administration of IVT brolocizumab (Beovu): ever; vamikibart: <=24 weeks prior to Day 1 in either eye. 9. Any proliferative diabetic retinopathy. 10. Active intraocular or periocular infection or active intraocular inflammation in the study eye. 11. Any current or history of ocular disease other than DME that may confound assessment of the macula or affect central vision in the study eye. 12. Any current ocular condition which, in the opinion of the investigator, is currently causing or could be expected to contribute to irreversible vision loss due to a cause other than DME in the study eye.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Vamikibart 1 mg administered in combination with ranibizumab 0.5 mg on Day 1 and every fourth week (Q4W) up to Week 44 (total of 12 injections). * **Route of Administration:** Intravitreal (IVT) injection to the study eye. * **Duration of Treatment:** 44 weeks of active treatment, followed by an observational period up to Week 72 (overall study duration: 76 weeks).

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard background medications for diabetes, hypertension, and other systemic co-morbidities. * **Prohibited:** Corticosteroid implants (e.g., Iluvien, Retisert within 3 years prior to Day 1), investigational ophthalmic agents, and macular laser therapies in the study eye.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Ophthalmic Exam, BCVA Assessment, OCT
Baseline	Day 1	Randomization, First IVT Injection (Vamikibart+Ranibizumab or Ranibizumab+Sham)
Interim	Q4W (Weeks 4-44)	IVT Injections, Safety Assessments, BCVA, Central Retinal Thickness
End of Study	Week 72 / 76	Final Visual Acuity Assessment, Exit Interview, Observational Follow-up End

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Change From Baseline in Best Corrected Visual Acuity (BCVA) Averaged Over Week 44 and Week 48 in Treatment-naïve Participants. * **Measurement Method:** BCVA was measured via Early Treatment Diabetic Retinopathy Study (ETDRS) chart at a starting distance of 4 meters using a set of three Precision vision or Lighthouse distance acuity charts (modified ETDRS Charts 1, 2, and R) prior to dilating eyes by a trained and certified VA examiner. The BCVA letter score ranges from 0 to 100 letters. Higher scores and gain in BCVA from baseline indicate improvement in VA. This analysis used a Mixed Model for Repeated Measurements (MMRM) model.

7.2 Secondary & Exploratory Endpoints * Number of Participants With Systemic and Ocular Adverse Events (AEs). * Change From Baseline in BCVA Averaged Over Week 44 and Week 48 in Previously Treated Participants. * Change From Baseline in BCVA Averaged Over Week 44 and Week 48 in Overall ITT Population. * Change from baseline in Central Subfield Thickness (CST) as assessed by Optical Coherence Tomography (OCT). * Proportion of patients achieving ≥ 15 -letter improvement or experiencing ≥ 15 -letter loss in BCVA. * Pharmacokinetics (PK) and pharmacodynamics (PD) of intravitreal vamikibart.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Routine collection via ophthalmic exams and patient reporting, with a focus on systemic and ocular AEs (e.g., intraocular inflammation, increased intraocular pressure, eye pain).

- **Serious Adverse Events (SAEs):** Required to be reported to the sponsor within 24 hours of site awareness.
 - **Data Monitoring Committee (DMC):** Independent external committee responsible for periodic safety data review.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) for primary efficacy outcomes; Safety Set for all subjects who receive at least one dose of the investigational product.
 - **Sample Size Justification:** Powered ($\alpha = 0.05$) to detect a statistically significant difference in BCVA letter score improvement between the combination arm and ranibizumab monotherapy.
 - **Handling of Missing Data:** Mixed-Model Repeated Measures (MMRM) and/or Multiple Imputation for continuous efficacy endpoints.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Scheduled onsite and remote monitoring visits to ensure protocol adherence, IVT masking integrity, and safety compliance.
 - **Audit Trail:** 21 CFR Part 11-compliant electronic Data Capture (eDC) system with integrated data clarification forms.
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11. Study Identifiers & Governance

- **Primary ID:** BP43464
 - **Registry Number:** NCT05151744
 - **Posting ID:** 535366253228634946
 - **Sponsor/Lead Organization:** Hoffmann-La Roche (Roche)
 - **Study Title:** A Study to Investigate Vamikibart (R07200220) in Combination With Ranibizumab in Diabetic Macular Edema
 - **Official Regulatory Title:** A Phase II, Multicenter, Randomized, Double Masked, Active Comparator-Controlled Study to Investigate the Efficacy, Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of R07200220 in Combination With Ranibizumab Administered Intravitreally in Patients With Diabetic Macular Edema
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12. Clinical Framework (Disease Specific)

- **Primary Condition:** Diabetic Macular Edema (DME) / Macular edema due to diabetes mellitus

- **Ontology Classifications:** MeSH: D000085890 | SNOMED CT: 312912001
 - **Diagnosis Threshold:** Decreased visual acuity and central macular thickening observed on OCT.
 - **Medical Methodology:** Standard anti-VEGF therapy combined with novel anti-IL-6 biological targeting.
 - **Optimization Target:** Maximal reduction of macular edema and optimal restoration of visual acuity.
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13. Study Procedures & Methodology

- **Management Pathway:** Intravitreal (IVT) injection protocol and continuous retinal architecture monitoring.
 - **Education Protocol (HCP):** Site training on masked IVT administration, standardized BCVA testing, and OCT capture protocols.
 - **Education Protocol (Patient):** Education on expected post-injection symptoms, signs of ocular infection (e.g., endophthalmitis), and diabetes management.
 - **Quality Audit Mechanism:** Central reading center utilized for independent, masked evaluation of OCT images and BCVA data quality assurance.
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14. Observation & Follow-Up Schedule

- **Total Duration:** 76 Weeks (44 weeks treatment + 32 weeks observation).
 - **Onsite Visit Cadence:** Every 4 weeks (Q4W) for the active phase.
 - **Remote Monitoring Frequency:** Follow-up contacts scheduled as per protocol during the observational period.
 - **Checklist Review Interval:** Regular safety and visual acuity checks Q4W.
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				